Compensation Analysis of Major League Baseball Players

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1 Introduction & Research Question

This work focuses on analyzing the compensation of Major League Baseball (MLB) players, with the primary objective of determining the extent to which athletic performance influences a player's annual salary.

The central problem consists of robustly modeling the relationship between performance statistics—whether single-season or career-wide—and salary, while handling the statistical complexities of real-world data.

> **To what extent are Major League Baseball players compensated based on their athletic performance?**

2 Presentation of the Reference Article

This study is based on the article *"A Critical Look at Some Analyses of Major League Baseball Salaries"* written in 1995 by Hoaglin and Velleman [1].

This article originated from a challenge issued to fifteen groups of statisticians: each group was tasked with analyzing the exact same dataset on baseball salaries to understand how they are determined. The authors' objective was not to uncover a single magic formula for salaries, but rather to examine how different experts approach real-world data and which statistical methods perform best.

The paper highlights both the strengths and the pitfalls of various analytical methods. Hoaglin and Velleman demonstrate that mathematical formulas should never be applied blindly. They emphasize the importance of thoroughly exploring the data beforehand, applying appropriate variable transformations when necessary, and accounting for extremely high salaries that can distort model estimates. In short, the article serves as a methodological guide for building a robust and reliable statistical model despite real-world data anomalies.

3 Presentation of the Dataset

The data used in this study originate from official American baseball records, combining 1987 annual salaries with athletic performance metrics from the 1986 season.

The dataset covers 322 players, primarily hitters, for whom extensive numerical metrics are recorded. These figures include recent single-season statistics, such as the number of hits or runs scored in 1986, as well as cumulative career statistics measuring consistency and league experience.

In addition to performance metrics, the dataset contains contextual information such as the player's team, field position, and league. One notable characteristic of these data is the wide disparity in compensation: most players earn moderate salaries, whereas a small minority

command extraordinarily high incomes. This skewness, combined with missing salary records for certain players, requires careful data cleaning and transformation prior to constructing a reliable regression model.

4 Descriptive Analysis & Results

4.1 Dataset Overview

The dataset comprises 28 variables, including 7 qualitative variables. After importing the data and removing rows with missing salary entries, we obtain a clean sample of $n = 263$ players, all of whom are hitters.

4.2 Univariate Analysis: Salary Distribution

Examining the target variable "Salary" (`Salary_1987`) is the pivotal step of this study. This preliminary analysis reveals a highly specific market structure that dictates our modeling strategy.

4.2.1 Observation: Pronounced Right-Skewness (Histogram)

Direct inspection of the raw distribution (see Figure 1) reveals a unimodal shape that is heavily skewed to the right:

- **Concentration:** The vast majority of observations are concentrated in the lower salary brackets.
- **Decay Profile:** The frequency of salaries decreases almost exponentially as compensation increases, creating a "long tail" toward upper values.
- **Consequence:** The mean (\$542.2k) is substantially higher than the median (\$430.0k). This gap confirms that a small number of extremely high salaries pull the distribution upward, distorting the perception of a "typical player".

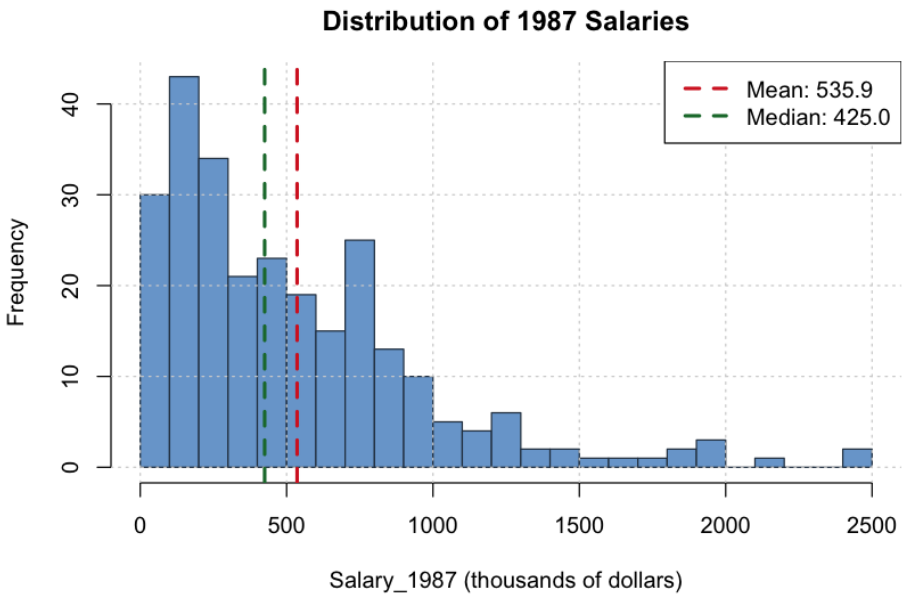


Figure 1: Histogram of 1987 salaries

4.2.2 Quartile Analysis and Outlier Detection (Boxplot)

Using a Boxplot (see Figure 2) allows us to mathematically segment the population and isolate atypical observations:

- **Core of the Market (Interquartile Range - IQR):**
  - Q_1 (25%): \$193,000.
  - Q_3 (75%): \$750,000.
  - **Interpretation:** 50% of players earn between these two thresholds. This interval represents the "middle class" of the league.
- **Outlier Threshold (Tukey's Method):**

$$\text{Threshold} = Q_3 + (1.5 \times \text{IQR}) = \$1,585,500$$

- **Result:** 11 extreme outliers are detected (approximately 3.4% of the initial dataset, or 4.2% of the 263 complete cases). These superstars earn salaries so high that they no longer follow the statistical pattern of the rest of the league. In an unadjusted linear model, they would exert disproportionate leverage and degrade predictive accuracy for the remaining players.



Figure 2: Boxplot of 1987 salaries

4.2.3 Logarithmic Transformation: A Corrective Necessity

To address these distributional biases, applying a logarithmic transformation (log(Salary)) achieves three objectives:

- **Symmetrization of the Distribution:** The logarithm compresses distances at the upper extreme. The resulting distribution approximates a normal distribution, an optimal condition for the validity of parametric statistical tests.

- **Variance Stabilization (Homoscedasticity):** In the raw data, dispersion tends to grow with salary level. The log transformation stabilizes this variance, ensuring the model maintains consistent precision across both bench players and stars.
- **Reduction of Outlier Leverage:** By bringing multi-million-dollar figures onto a comparable scale, the 11 outliers are reintegrated into the main body of the distribution without discarding the information they carry.

4.2.4 Toward Coherent Linear Modeling

The ultimate goal is to prepare the response variable for linear regression modeling (see Figure 3):

- **Paradigm Shift:** Without the logarithm, the model seeks an additive relationship (e.g., "+1 Hit = +\$500"). With the logarithmic transformation, the model becomes multiplicative. We no longer analyze raw dollar increments, but rather relative percentage changes.
- **Elasticity:** This allows salary to be explained by performance metrics (Hits, Home Runs, etc.) in terms of elasticity (e.g., "+10% in Runs → +5.16% in salary"). This formulation aligns much more closely with the economic reality of professional sports, where differences in talent are rewarded exponentially.

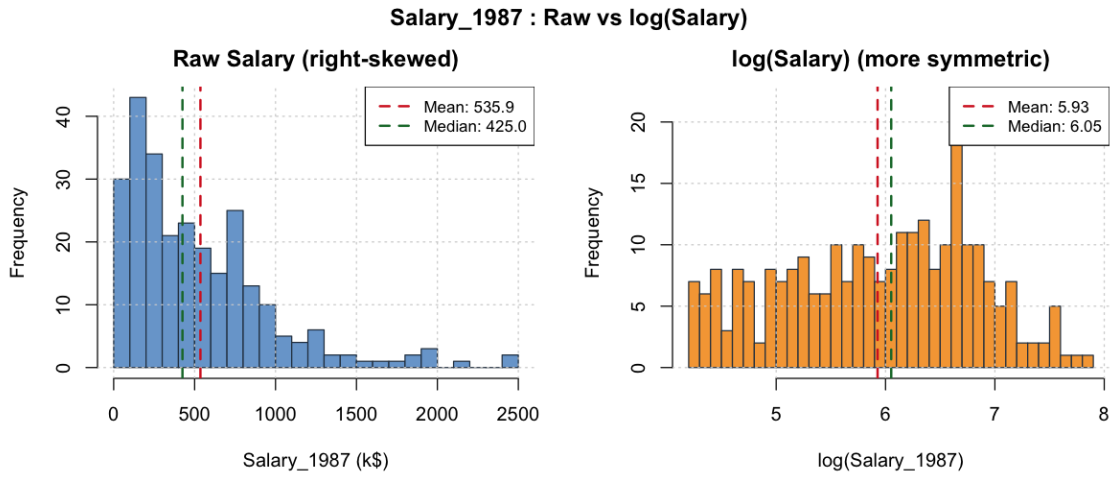


Figure 3: Distribution of raw salaries and after logarithmic transformation

5 Econometric Modeling & Results

As the descriptive analysis revealed significant correlations between log(Salary) and several performance metrics, three predictors were selected for modeling: career duration (`Longevity`), number of hits in 1986 (`Hits_86`), and an engineered feature, `Hits_per_year`, obtained by dividing total career hits by years of experience:

$$\text{Hits per year} = \frac{\text{Career Hits}}{\text{Longevity}}$$

This normalization is necessary because raw career totals mechanically favor veteran players: 500 hits over 3 years is not equivalent to 500 hits over 15 years. `Hits_per_year` thus measures average annual productivity independently of seniority. These three variables were first evaluated individually in simple regressions and then combined into a multiple regression model.

5.1 Simple Regressions

Three simple linear regressions were fitted to isolate the individual effect of each predictor (see Figure 4). Model M1 examines the impact of experience alone on salary, M2 evaluates recent single-season performance, and M3 assesses average annual performance over a player's career:

$$\begin{aligned}
\text{M1:} \quad & \log(\text{Salary}) = 5.173 + 0.1047 \times \text{Longevity} \quad (R^2 = 0.322) \\
\text{M2:} \quad & \log(\text{Salary}) = 4.821 + 0.0103 \times \text{Hits_86} \quad (R^2 = 0.263) \\
\text{M3:} \quad & \log(\text{Salary}) = 4.572 + 0.0151 \times \text{Hits_per_year} \quad (R^2 = 0.430)
\end{aligned}$$

All three models are statistically significant. M1 indicates that an additional year of experience is associated with an approximate 11% increase in salary ($e^{0.1047} \approx 1.11$). M3 achieves the highest R^2 among the three (0.430), suggesting that average career performance explains salary better than single-season output. Taken individually, however, these models remain insufficient: the strongest single predictor accounts for only 43.0% of salary variance, motivating the construction of a combined multiple regression model.

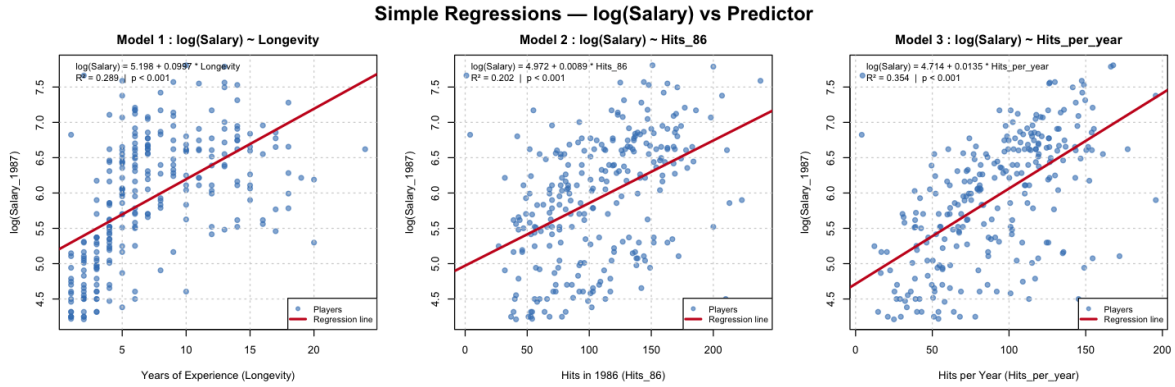


Figure 4: Relationship between log(Salary) and each predictor (M1, M2, M3)

5.2 Multiple Regression

To combine the three dimensions identified above, a multiple linear regression model (M4) was constructed by simultaneously incorporating `Longevity`, `Hits_per_year`, and `Hits_86`:

$$\log(\text{Salary}) = 3.993 + 0.0875 \times \text{Longevity} + 0.0065 \times \text{Hits_per_year} + 0.0066 \times \text{Hits_86}$$

All three predictors in Model M4 are statistically significant ($p < 0.001$). The interpretation of the estimated coefficients is as follows:

- `Longevity` ($\beta = 0.0875$)**:** Each additional year of service corresponds to an approximate +9.1% increase in annual compensation ($e^{0.0875} - 1$), holding other variables constant.
- `Hits_per_year` ($\beta = 0.0065$)**:** Each additional hit per career year is associated with an approximate +0.6% increase in salary.
- `Hits_86` ($\beta = 0.0066$)**:** Each additional hit during the 1986 season is associated with an approximate +0.7% increase in salary.

Longevity remains the dominant predictor, exhibiting a substantially larger effect size than the two hitting metrics, which suggests that league tenure is rewarded more heavily than raw hitting output alone in this specification (see Figure 5).

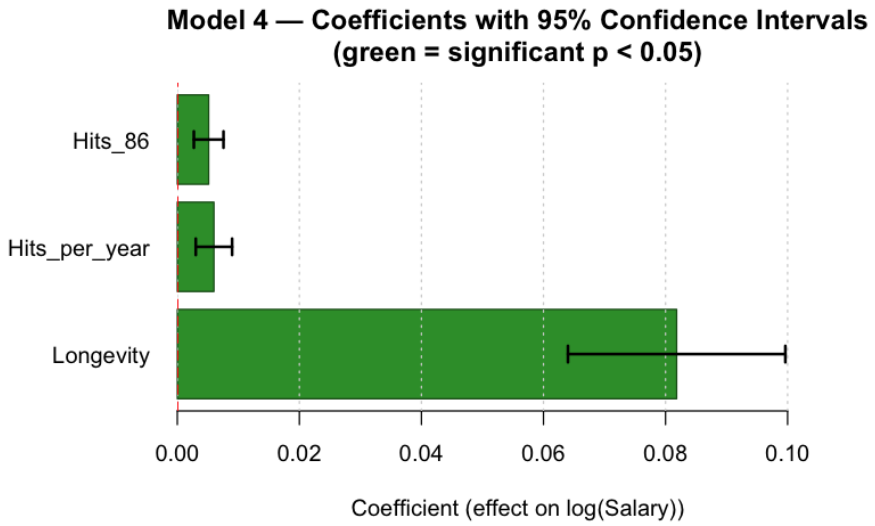


Figure 5: Model M4 coefficients with 95% confidence intervals

5.3 Goodness of Fit

Model M4 achieves an R^2 of 0.621 ($R^2_{\text{adj}} = 0.617$), representing a substantial improvement over the univariate specifications. The adjusted R^2, which penalizes the inclusion of uninformative predictors, remains nearly identical to the unadjusted R^2, confirming that each of the three variables contributes genuinely to explanatory power. Table 1 summarizes the comparison across specifications:

Model	**Predictors**	R^2	**Adjusted** R^2
M1	`Longevity`	0.322	0.319
M2	`Hits_86`	0.263	0.260
M3	`Hits_per_year`	0.430	0.428
M4	Combined	**0.621**	**0.617**

Table 1: Comparison of regression models

M4 accounts for 62.1% of the variance in log(Salary), compared to 43.0% for the best single-predictor model. The predicted-versus-actual plot (see Figure 6) visually confirms this fit: observations cluster consistently around the $y = x$ reference line, with residual dispersion expected for empirical salary data. These findings indicate that athletic performance, combined with league experience, explains a substantial share of compensation disparities.

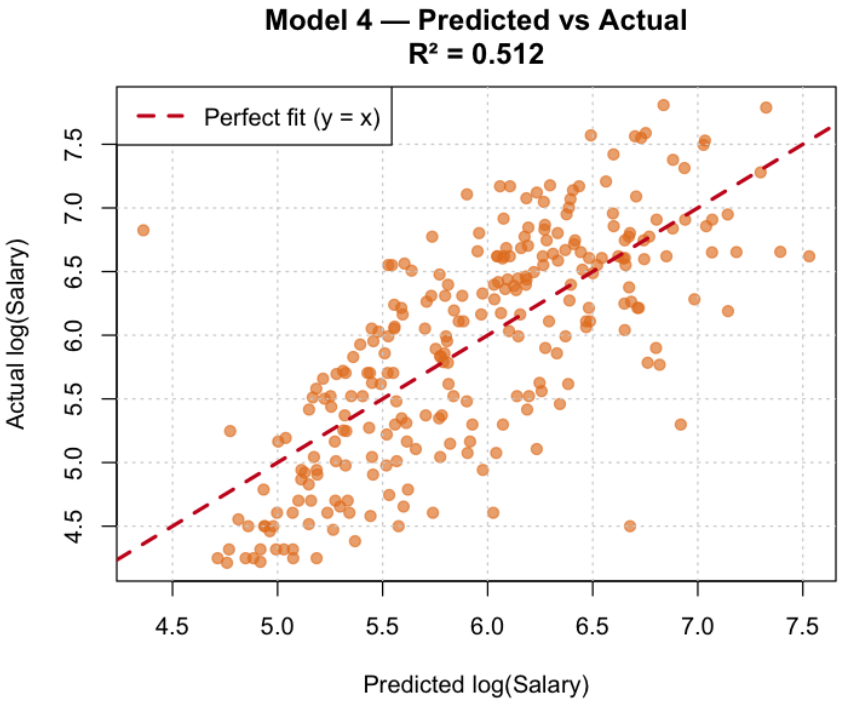


Figure 6: Predicted vs. actual values for Model M4 ($R^2 = 0.621$)

6 Tests & Diagnostics

The diagnostic checks presented in this section verify that the regression model rests on sound statistical foundations. Two core assumptions are evaluated: the normality of residuals and the absence of systematic structure in the error terms.

6.1 Normality of Residuals

After fitting a model, each player has an associated residual: the difference between actual log-salary and the value predicted by the model. For inferential statistics to remain reliable, these residuals must behave randomly and follow an approximately normal distribution. Two tools are used to verify this property.

The Normal Q-Q plot compares the empirical distribution of residuals against theoretical normal quantiles. If the points align along the red reference line, the residuals are normally distributed and model inference is valid; deviations at the tails indicate skewness. The Q-Q plots across all four models (see Figure 7) show that M1 and M4 closely track the diagonal across the majority of their range, whereas M2 and M3 exhibit sharper deviations in the lower tails, revealing slight negative skewness.

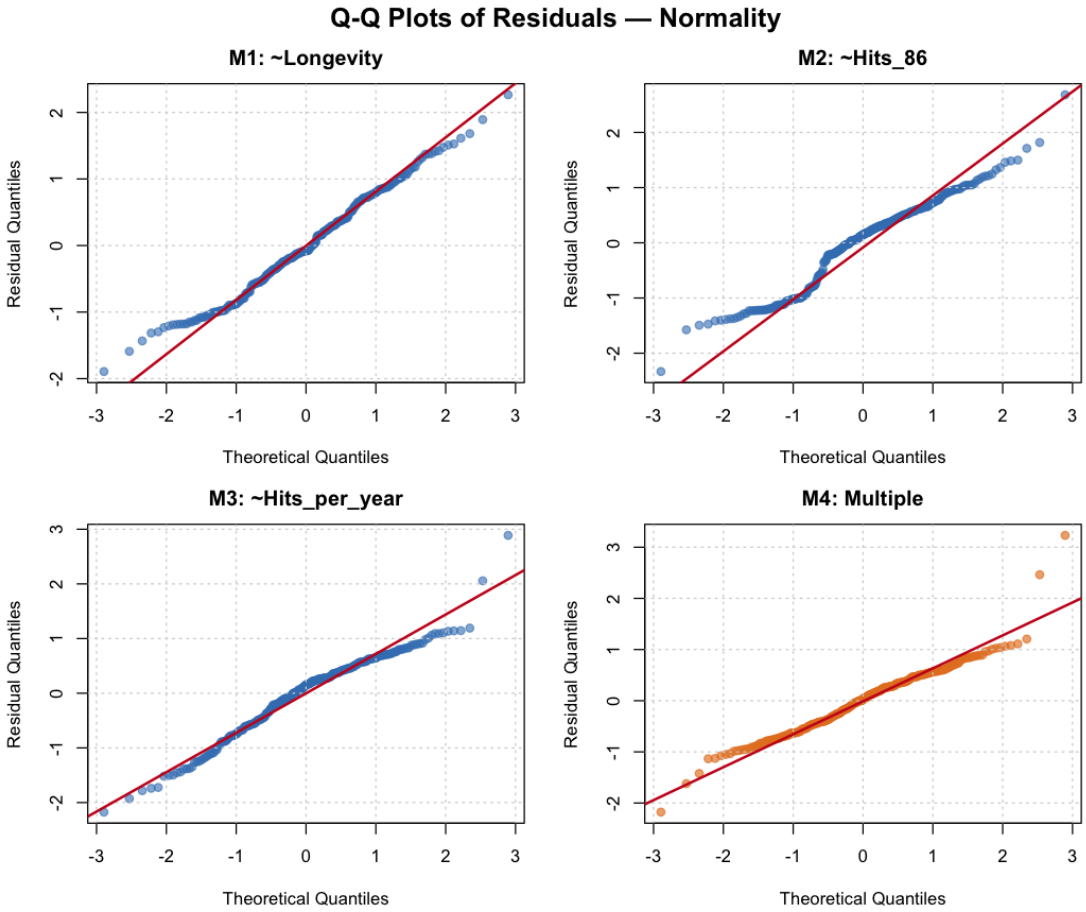


Figure 7: Q-Q plots of residuals across all four models

The Shapiro-Wilk test formalizes this check: the null hypothesis H_0 states that the residuals are normally distributed. Applied to the residuals of M4, the test yields $W = 0.9878$ with $p = 0.0248$. H_0 is formally rejected at the $\alpha = 5\%$ significance level, indicating a statistically detectable departure from exact normality. However, $W = 0.9878$ is remarkably close to 1 (the value corresponding to exact normality), indicating that the practical magnitude of this deviation is negligible. With $n = 263$ observations, the Shapiro-Wilk test becomes highly sensitive to minor tail discrepancies.

6.2 Homoscedasticity and Residual Structure

The residuals-versus-fitted-values plot (see Figure 8) detects whether the model commits systematic specification errors. A well-specified model yields residuals scattered randomly around zero without discernible patterns. The presence of a pronounced curve, arch, or funnel indicates unmodeled structure in the data.

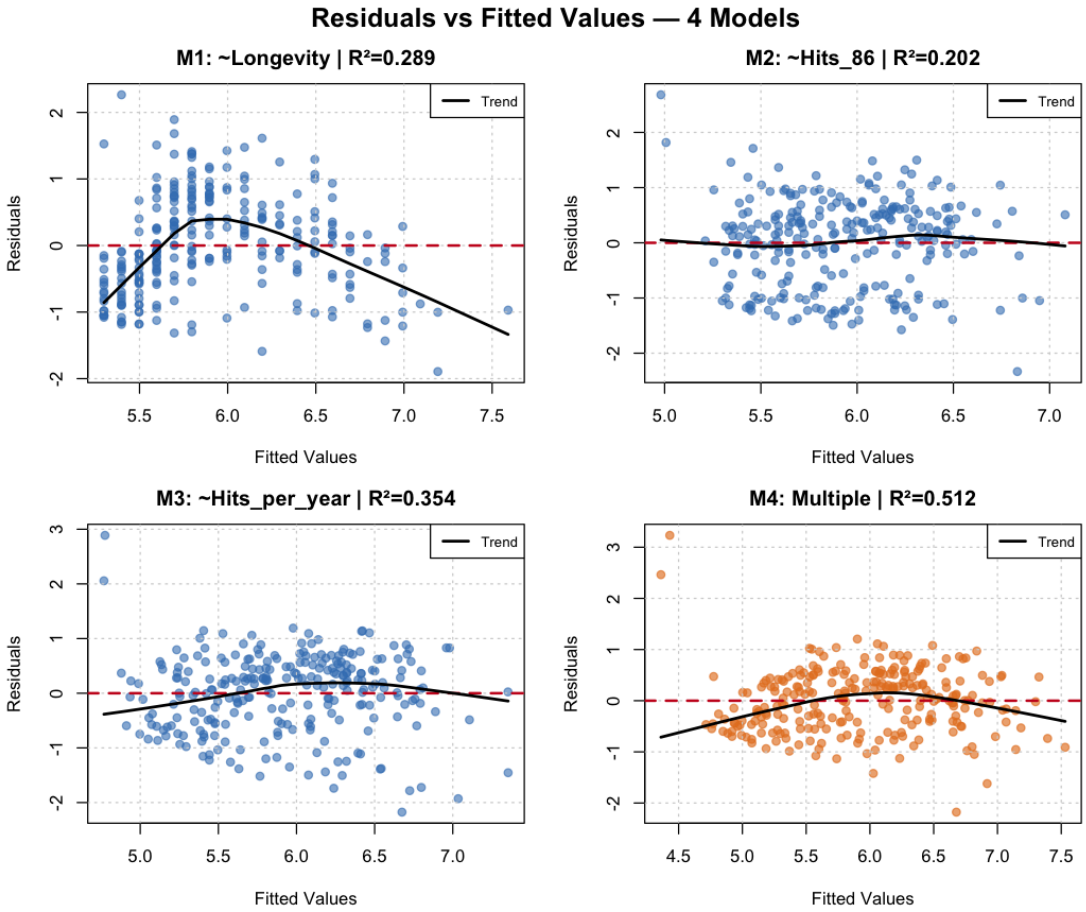


Figure 8: Residuals vs. fitted values across all four models

M1 displays a pronounced arch shape: the model underestimates mid-career salaries and overestimates veteran salaries, confirming that the relationship between `Longevity` and log(Salary) is non-linear. M2 exhibits the flattest trend curve around zero among the simple models, but its low R^2 of 0.263 shows that residual linearity does not compensate for weak explanatory power. M3 and M4 behave similarly, showing only a slight downward drift at the upper extreme without major systematic structure across the fitted range.

6.3 Conclusion on Model Validity

The diagnostics confirm that M4 is the most reliable specification. The minor departure from normality detected by the Shapiro-Wilk test is negligible in practice and does not invalidate statistical inference. The absence of severe systematic structure in the residuals of M4 validates its use for answering our research question.

7 Discussion & Limitations

The empirical findings provide concrete answers to our research question. Model M4 explains 62.1% of salary variance, representing strong explanatory power for real-world compensation data. Both athletic performance and league experience contribute significantly to player salaries. We can therefore conclude that players are **partially** compensated according to their performance, though performance is not the sole determinant.

The remaining 37.9% of unexplained variance reflects structural factors omitted from the model. Fielding position is not accounted for: a catcher and an outfielder do not share the same

salary structure even at identical batting output. Franchise payroll budgets also play a critical role, as large-market franchises possess far greater financial resources than small-market teams regardless of individual player metrics. Contractual factors such as free-agency eligibility and salary arbitration clauses directly shape contract negotiations without appearing in performance tables. Furthermore, excluding the 59 observations with missing `Salary_1987` entries reduced the sample from 322 to 263 players, which could introduce selection bias if missingness is not completely at random.

Methodologically, our diagnostics demonstrated that the relationship between `Longevity` and log(Salary) is not strictly linear, as evidenced by the arched residual pattern in M1. Our specification models this effect linearly as a simplifying approximation. Hoaglin and Velleman (1995) suggested introducing a quadratic term for `Longevity` to better capture this non-linearity, an extension left for future work.

8 Conclusion

Exploratory data analysis revealed a heavily right-skewed salary distribution, motivating the use of a variance-stabilizing logarithmic transformation. Simple linear regressions confirmed that career longevity, 1986 hits, and career average hits per year are each significantly associated with compensation. The multiple regression specification M4, combining all three predictors, achieves an R^2 of 0.621 (explaining 62.1% of log-salary variance), supported by satisfactory residual diagnostics.

These results answer our research question affirmatively yet with nuance: athletic performance does significantly drive Major League Baseball compensation, but it is not the exclusive factor. Career longevity, reflecting accumulated experience and tenure rather than pure hitting efficiency, emerges as the dominant predictor. The 37.9% of unexplained variance underscores the role of unmodeled institutional determinants such as defensive position, franchise payroll capacity, and contract negotiation status.

References

[1] Hoaglin, D. C., & Velleman, P. F. (1995). *A Critical Look at Some Analyses of Major League Baseball Salaries.*